

Figure 1

A vision-language agentic pipeline for lead optimization

Twelve deterministic stages map a lead structure and its ligand-interaction diagram to a ranked set of synthetically-accessible analogs. Three substrate categories — upstream open-source primitives, Benchside-implemented logic, and defence-in-depth gates — are visually distinguished. Stage indices match backend/app/services/lead_optimizer/orchestrator.py.



9–10

Pre-filter + ADMET screen

Lipinski · PAINS · Brenk · admet_ai re-prediction

Hard structural filters drop invalid valencies, PAINS, Brenk alerts, and ADMET-failing analogs (directional scoring: hERG / Skin / CYP risk vs HIA / BBB benefit).

PAINS

Brenk

directional

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Synth-accessibility

SYBA (Voršilák 2020, Bayesian) · Ertl SAScore (1–10) fallback

SYBA signed score (~–50 easy to ~+50 hard) is the platform-ratified synth-accessibility metric. Sigmoid-centered penalty feeds into ranking.

SYBA

SAScore

12

Ranking and report

Weighted scalar sum × penalty multiplier · WeasyPrint PDF / HTML / CSV

Single-score "Pareto-style penalty rank" — sorted total_score over docking, ADMET, SYBA, and structural axes. Top-K written to PDF/HTML/CSV with per-analog SMILES, modifications, SYBA verdict, ADMET profile, and structure depiction.

SYBA-anchored

multi-format

out

Outputs

Ranked analog list · per-analog ADMET + SYBA dossier · PDF / HTML / CSV reports · email delivery

¹ **Per-instance labelling.** Two phenyl rings on the same molecule would collapse to a single generic SMARTS (c1ccccc1) without disambiguation. The pipeline derives left/right/top/bottom position hints from RDKit-generated 2D coordinates and emits atom-anchored SMARTS via `MolFragmentToSmarts`, so each instance carries a distinct restricted/editable verdict.

² **Soft Murcko gate, where it actually lives.** The Bemis–Murcko scaffold SMARTS is appended to the restricted-SMARTS set at *stage* 3. The gate itself is implemented by `enforce_pharmacophore()`, called twice — once at *strategy validation* (does this SMIRKS, applied to one example, preserve the pharmacophore?) and once for every analog during *permutation*. Predicates checked: (a) restricted atoms unchanged, (b) ring-system topology preserved, (c) aromaticity preserved. Allows topologically equivalent edits (methyl→ethyl on a scaffold carbon); rejects ring destruction.

³ **Chemistry validator coverage.** Allowlist tables for H-bond donor, H-bond acceptor, π -stack, hydrophobic, salt bridge, and cation- π . Each entry maps a functional group family to its allowed interaction types. Vision-model output that violates the chemistry tables is dropped before SMARTS construction (stage 3), so a methyl group cannot enter the pipeline as a hydrogen-bond donor.

⁴ **SMIRKS library scope.** 479 curated SMIRKS transformations across 22 categories (aromatic-ring swaps, amine modifications, carbonyl modifications, CNS penetration, nitrogen-heterocycle swaps, polarity adjustments, halogen substitutions, steric shielding, metabolic stability, etc.) in the version inventoried for this paper. All entries carry a `validated=True` flag, but validation is curator-asserted, not gated by an automated RDKit-reaction-fires-on-target CI test — known limitation, fix planned before the supplementary release.

⁵ **Ranking metric.** The `total_score` is a weighted scalar sum across docking, ADMET liabilities, SYBA penalty, and structural alerts, multiplied by hard-filter penalties. The output is single-sorted; the field label "Pareto" is API-convention rather than a Pareto front. Honest framing in the paper: *weighted scalar penalty rank with SYBA as the synth-accessibility anchor*.

Benchside lead-optimization workbench, code-grounded as of 2026-06